LabTrack - Database Design

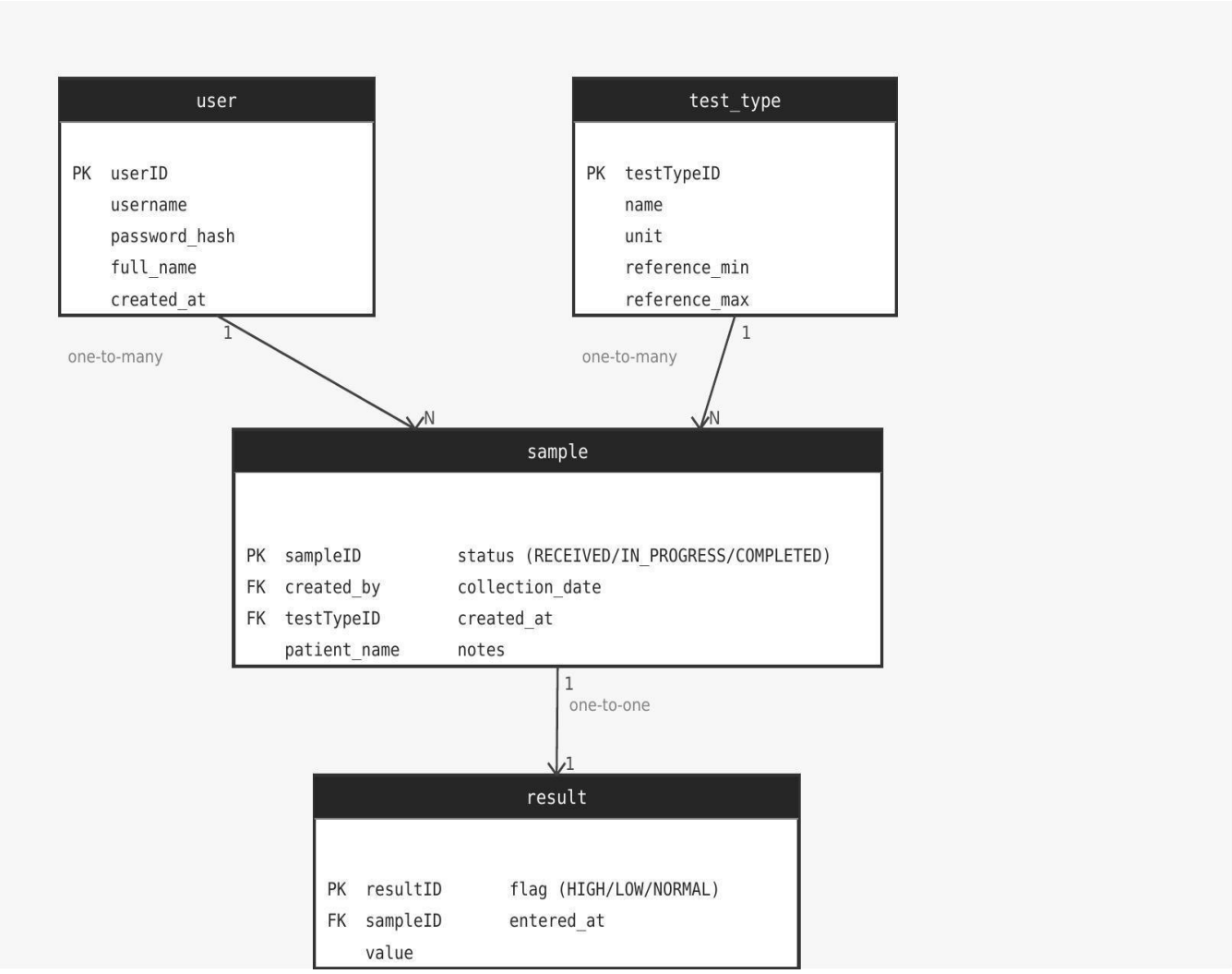
1. Database Technology Choice

I chose MySQL 8 for this project. The data naturally fits a relational structure. A sample belongs to a user and belongs to a test type and has one result. Those connections are straightforward foreign keys and exactly what MySQL is good at.

I thought about MongoDB but it did not make sense here. The reference ranges are stored in the test_type table and shared across many samples. If I embedded them in each sample document, changing a reference range would mean updating every single sample. MySQL handles that with one row update.

MySQL also works well for deployment. Railway and Render both offer free managed MySQL instances that connect straight to Spring Boot and Spring Data JPA. Database credentials will be stored as environment variables, not in the code.

2. Entity-Relationship Diagram



3. Table Definitions

3.1 users

Stores login credentials for everyone who uses the app. When Sam opens LabTrack and enters her username and password, Spring Security checks this table to verify her identity. Her user_id then gets stamped onto every sample she registers as the created_by value, so there is always a record of who registered what.

Column	Type	Constraint	Notes
user_id	INT	PK, AUTO_INCREMENT	Unique account ID
username	VARCHAR(50)	NOT NULL, UNIQUE	Login username
password_hash	VARCHAR(255)	NOT NULL	BCrypt hash
full_name	VARCHAR(100)	NOT NULL	Shown in navbar and on sample records
created_at	DATETIME	DEFAULT NOW()	Account creation time

3.2 test_types

Holds the catalog of tests the lab runs. When Sam selects a test type on the New Sample form, the test_type_id from this table gets stored on the sample. Later, when she enters the result, the service layer looks up reference_min and reference_max from this table and uses them to decide whether to flag the result as HIGH, LOW, or NORMAL. Pre-loaded with seed data on first deployment.

Column	Type	Constraint	Notes
test_type_id	INT	PK, AUTO_INCREMENT	Unique test type ID
name	VARCHAR(100)	NOT NULL, UNIQUE	e.g. Glucose, CBC, Lipid Panel
unit	VARCHAR(30)	NOT NULL	e.g. mg/dL
reference_min	DECIMAL(8,2)	NOT NULL	Low end of normal range
reference_max	DECIMAL(8,2)	NOT NULL	High end of normal range

Example: Glucose has reference_min=70, reference_max=140. A result of 185 exceeds 140, so the service layer assigns the flag HIGH.

3.3 samples

The main table. Every time Sam registers a new patient sample, a row is inserted here. The status column is what the workflow runs on. It can only go forward from RECEIVED to IN_PROGRESS to COMPLETED. The service layer blocks anything else.

Column	Type	Constraint	Notes
sample_id	INT	PK, AUTO_INCREMENT	Shown as S-YYYY-NNN

patient_name	VARCHAR(150)	NOT NULL	Patient full name
collection_date	DATE	NOT NULL	Date sample was collected
status	ENUM	DEFAULT 'RECEIVED'	RECEIVED, IN_PROGRESS, or COMPLETED
notes	TEXT	NULLABLE	Optional technician notes
created_by	INT	NOT NULL, FK → users	Who registered the sample
test_type_id	INT	NOT NULL, FK → test_types	What test it is for
created_at	DATETIME	DEFAULT NOW()	When it was registered
updated_at	DATETIME	DEFAULT NOW() ON UPDATE	Last update time

3.4 results

Stores the test result for each completed sample. When Sam enters a numeric result on the sample detail page, the service layer calculates the flag and writes both the value and the flag into this table. One result per sample only and enforced by the UNIQUE constraint on sample_id. Dr. Peter sees the flag immediately on his dashboard without anyone having to tag it manually.

Column	Type	Constraint	Notes
result_id	INT	PK, AUTO_INCREMENT	Unique result ID
sample_id	INT	NOT NULL, UNIQUE, FK → samples	One result per sample
value	DECIMAL(8,2)	NOT NULL	Numeric result entered by tech
flag	ENUM	NOT NULL	HIGH, LOW, or NORMAL – these are auto-set
entered_at	DATETIME	DEFAULT NOW()	When result was submitted

Example: value=185, flag='HIGH' because 185 > Glucose max of 140. This shows as a badge on the dashboard so Dr. Peter can spot it immediately.

4. Relationships

Tables	Type	Description
users → samples	One-to-Many	One user can register many samples.
test_types → samples	One-to-Many	One test type can apply to many samples.
samples → results	One-to-One	Each sample has at most one result. UNIQUE on sample_id enforces this.

5. Database Accounts

Account	Permissions	Purpose
labtrack_app	SELECT, INSERT, UPDATE	Runtime account used by Spring Boot. No schema change permissions.
labtrack_admin	Full privileges	Setup only. Not used by the running application.